

Assignment 3 - arrays and functions

100%

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Poengsum frakoblet modus:

100%



Legg til kommentar

Anonym vurdering: **nei**

15.12.2023

▼ Detaljer

Purpose

This assignment focuses on two new building blocks: Arrays and functions. The program you write will use variables and loops from last week together with these two new features.

Note: Make sure you learn the basics from the two lectures and lecture notes before attempting to solve this assignment. Also see the examples in Bitbucket.

Assignment 3.1

1. Create the project

Directory name: assignment_3_1

Create a CLion-project inside the `ikt101g23h/assignments/solutions/assignment_3_1` directory, as you did on the previous assignments.

2. Code your program

Write a program that reads in 10 integers, then (*The numbers here are just an example*):

- Prints out the lowest value:
Minimum: 5
- Prints out the highest value:
Maximum: 12
- Prints out the sum of all the values
Sum: 78
- Prints out the average (make sure you use %g with printf to round properly):
Average: 15.6
- Prints out the numbers, sorted
Sorted: 1 4 5
- Prints out the median
Median: 5.5

Requirements: Minimum, maximum, sum, average and median must be calculated using separate functions you write, not directly in main. Here how the sum function could look like (more in the lecture):

```
int sum(int numbers[], int count)
{
    ...
}
```

The functions must be called:

- *minimum* with return type *int*
- *maximum* with return type *int*
- *sum* with return type *int*
- *average* with return type *float*
- *median* with return type *float*

Approach: Try to work on one part at a time, same as last week. I recommend starting with reading in ten numbers and printing them back to make sure that part is working before you continue. Then write functions for each calculation (sum etc.) one at a time. Don't start on multiple things at once, build often, test often!

Tip: For the sorting have a look at `qsort`, which already exists in the standard library. You should be able to find an example that use it easily. If you want to sort manually then bubble sort might be the easiest.

3. Test locally

see *Local testing with Docker* under *Assignments*.

Test the solution on your own computer. You can fix your code and retest as many times as you want.

4. Deliver and test in Bamboo

See *Delivery and Bamboo testing* under *Assignments*.

Finally deliver and test the solution in Bamboo. This is what updates your score in Canvas which is the score I can see. You can test as many times as you want here as well up until the final deadline.